

WHY TDF COVERAGE IS LESS THAN SA?

REASON 1 :

Scenarios where 1 input of AND gate is tied to ground.

No transition faults were detected. But in SA, we were able to detect 2 faults

WHY TDF COVERAGE IS LESS THAN SA? (Contd...)

REASON 2 :

Because of timing exceptions

- False path

AU.FP : Faults untestable due to false path

- Multicycle path

AU.MCP : Faults untestable due to multicycle path

The information of false path and multicycle path will be written in SDC file
(Synopsys Design Constraints)

REASON 2 (Contd...) :

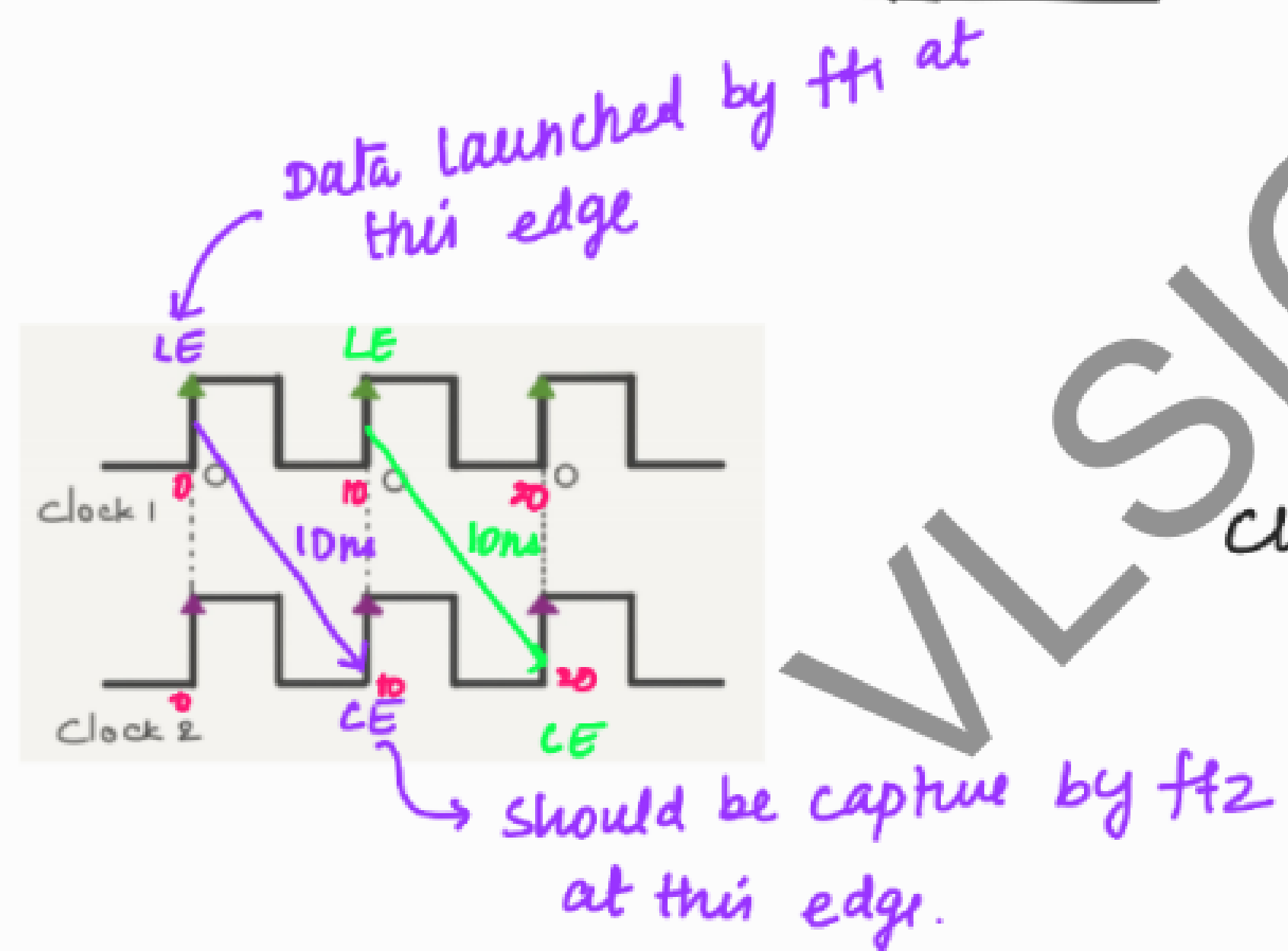
FALSE PATH :

For asynchronous path, designer will be declaring as false path so that the timing engine will not try to time the path.

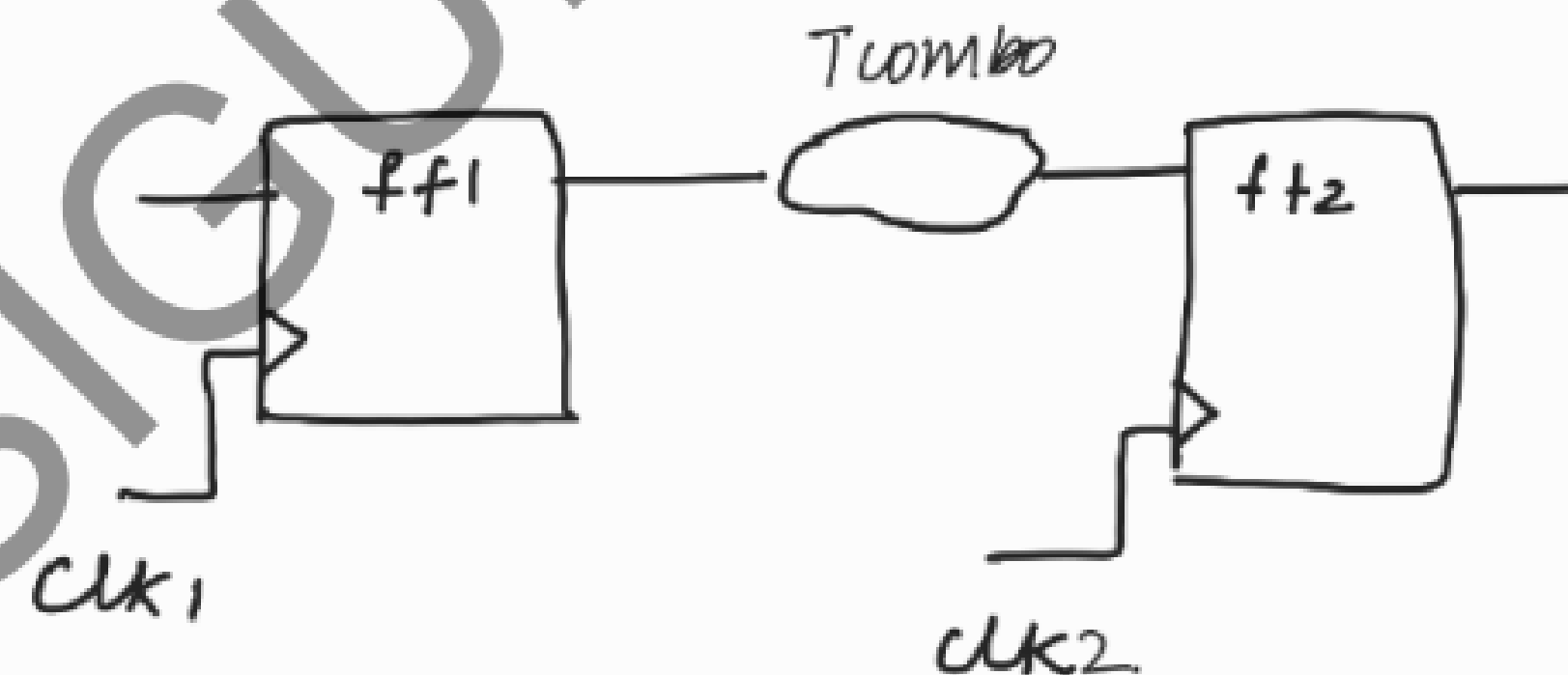
FALSE PATH :-

For asynchronous paths, designer will be declaring as false path so that the timing engine will not time those paths.

REASON FOR THE ABOVE STATEMENT:-



clock1 and clock2 are synchronous

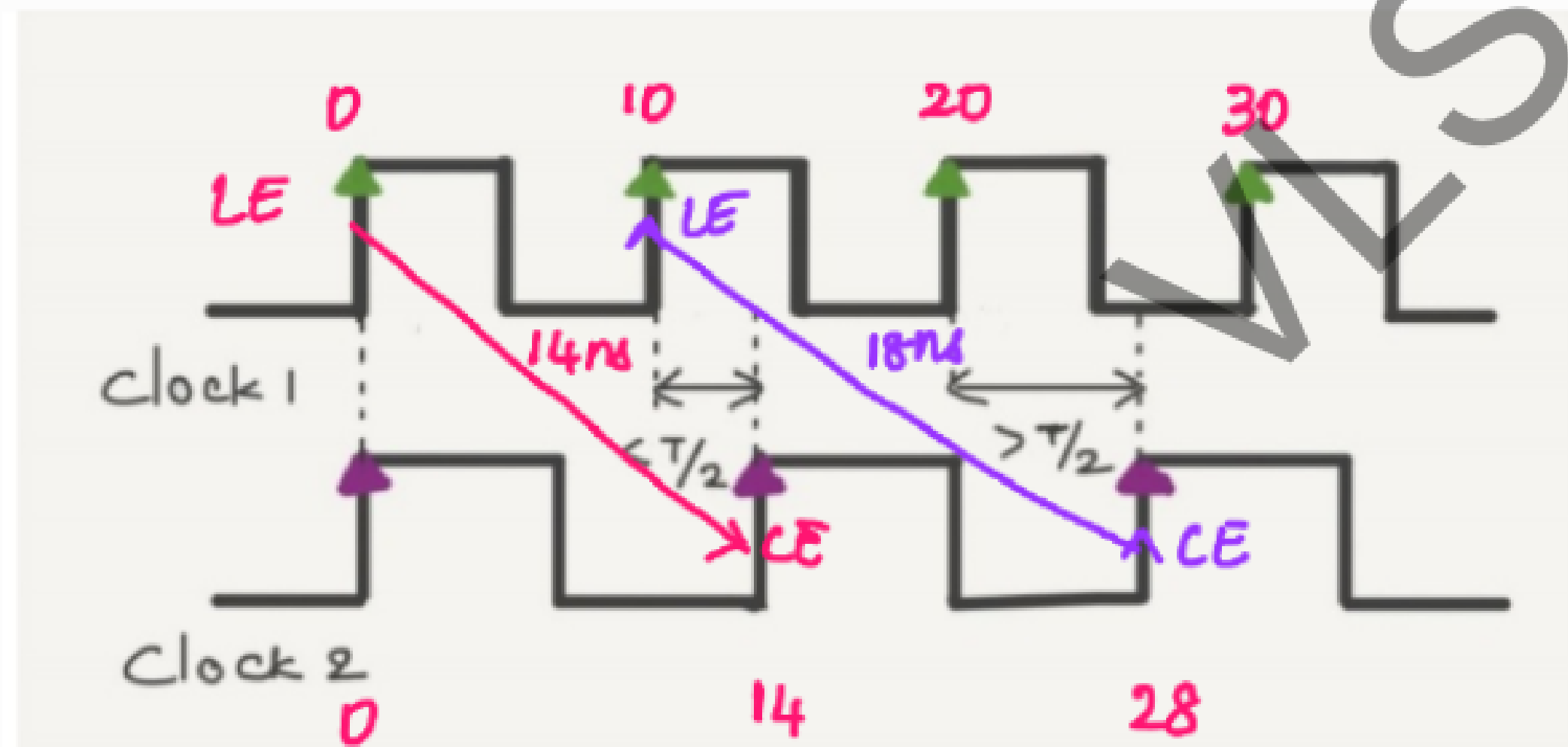


Suppose if 2 clks are synchronous, timing can be met as the exact gap b/w launch and capture edge is known.

But in the case of asynchronous clocks, we don't know the exact gap b/w LE and CE. It varies in each cycle.

So functional team can't meet the timing.

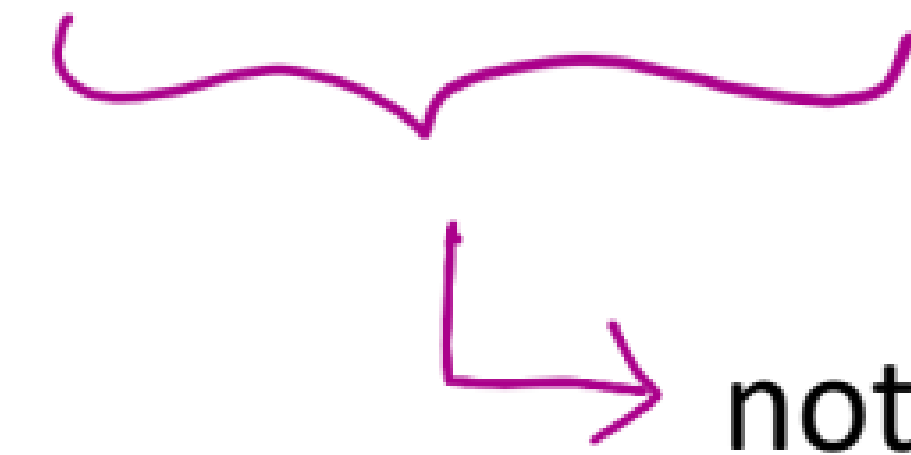
So designers declare asynchronous paths as false paths.



clock 1 and clock 2 are asynchronous

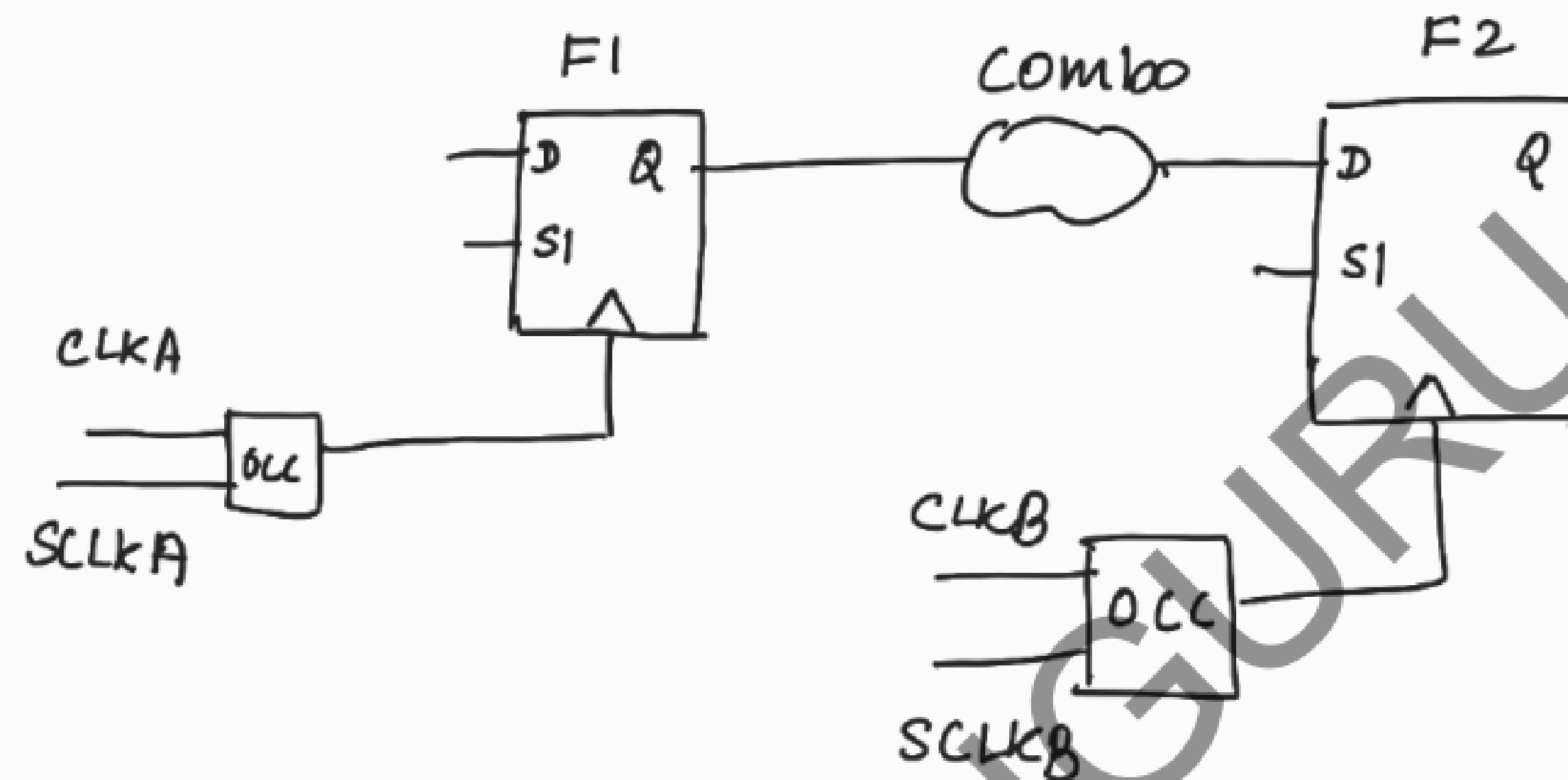
DFT engineers also cannot detect the TDF faults in these paths because :

TDF fault model is used to detect that the logic level from 0-1 and 1-0 properly transit within the specific time or not.



not known

How false paths are taken care during SA?

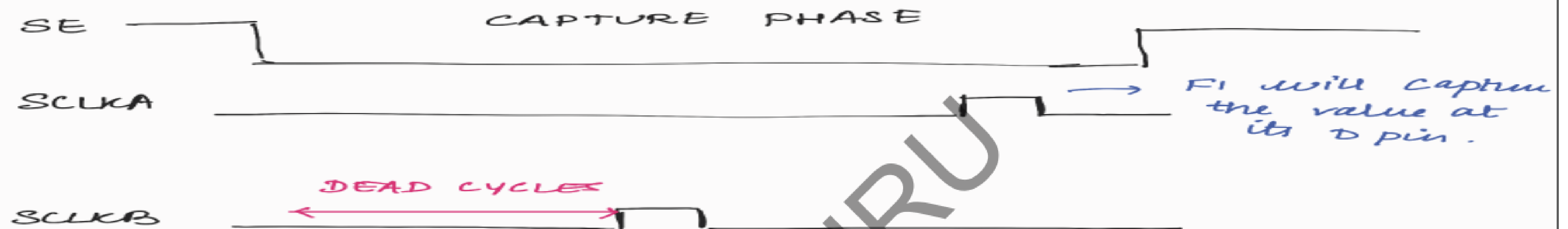


If CLK_A and CLK_B are asynchronous, designer will declare it as false path so that timing tool will not time those paths.

SCLK_A and SCLK_B will also be asynchronous.

As designer has declared it as a false path timing will not be closed for F1Q → F2D path.

For SA, tool will take care of the problem of not closing timing along that path.



The last shifted value of F_1 through the combo would have reached the D input of F_2 and would have got stabilized by the time this pulse comes as we can give dead cycles. The problem of setup time not met for the path F_1Q to F_2D is taken care now. When $SCLKB$ is pulsed, $SCLKA$ will not be pulsed. So F_2 will capture only last shifted value of F_1 through combo. There is no chance that F_2 will capture the newly captured value of F_1 as $SCLKA$ is not pulsed so F_1 will be retaining its last shifted value only. The problem of hold time not met for this path is also taken care now.

Note:

usually for these paths designer will be adding synchronizers

Synchronizers concept is not the scope of DFT engineers.

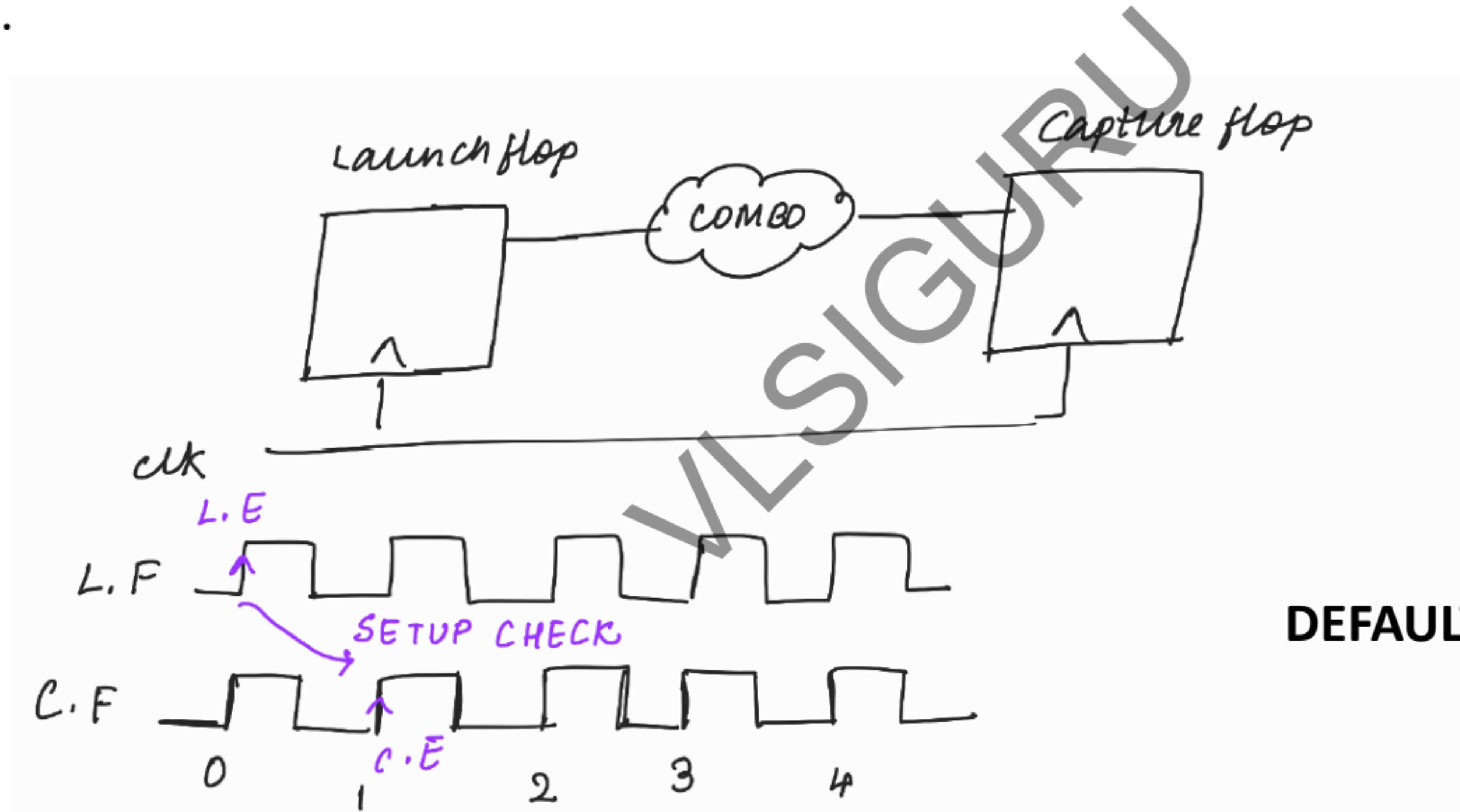
So I have just explained the concept without synchronizers.

Even if synchronizers are added, TDF will not be covered and SA will automatically be taken care by the tool.

REASON 2 (Contd...) :

MULTICYCLE PATH

Multicycle path is an exception of default single cycle timing requirement path. That is on MCP, signal requires more than one clock pulse to propagate the data from start point to end point.

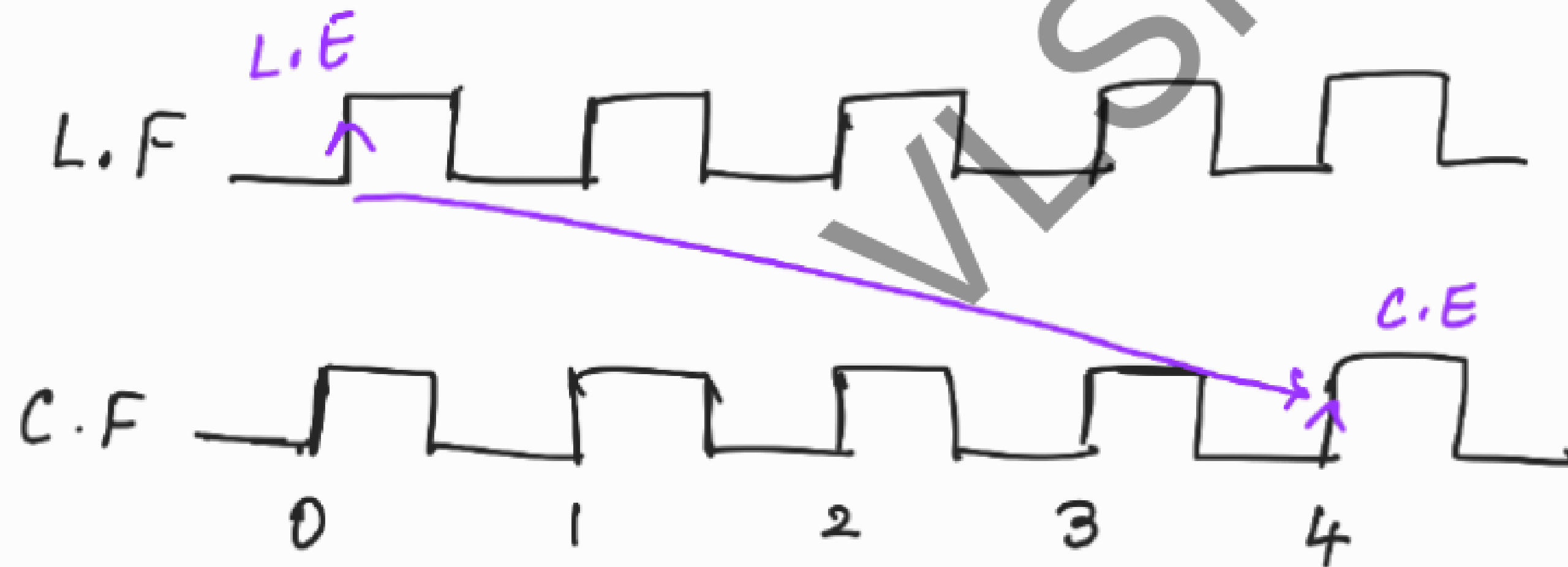


DEFAULT SCENARIO

Multicycle path scenario:-

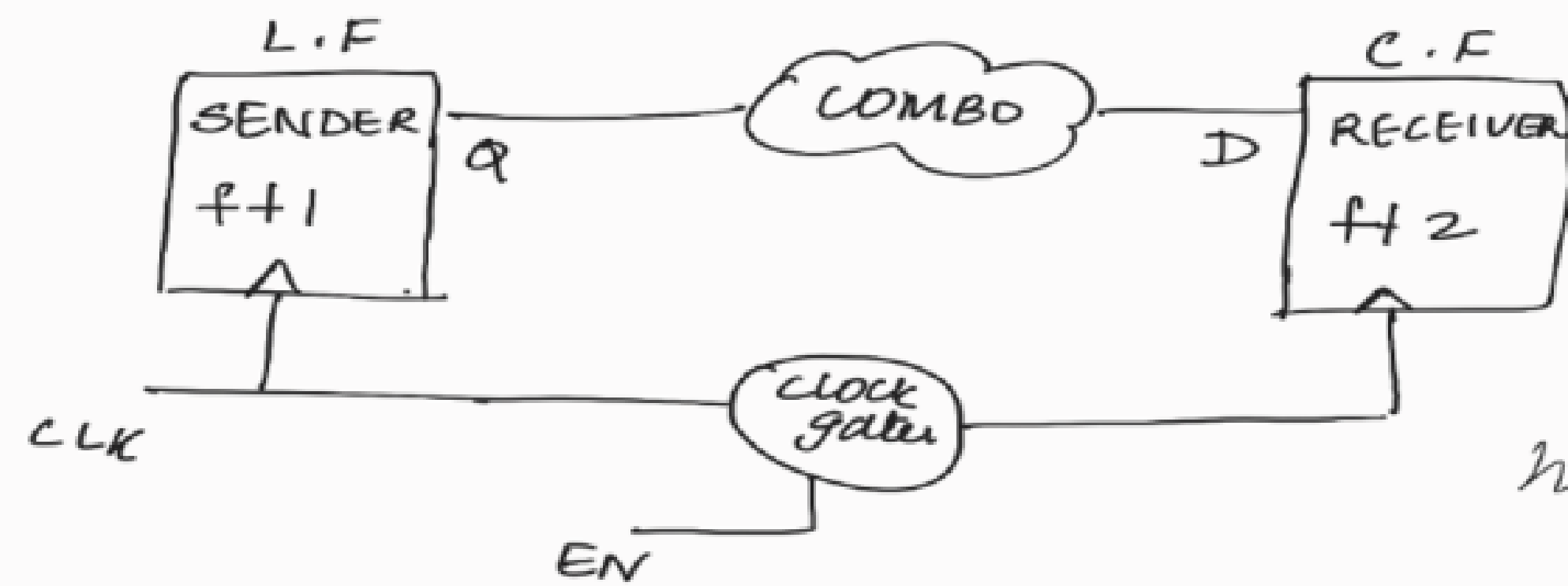
Sometimes the designer want to relax the setup check.

Instiad of checking at cycle 1, designer wants to check at cycle 4.



Design Scenario:

Eg:- If there is some situation like this in the design:- Triombo is very big. So the

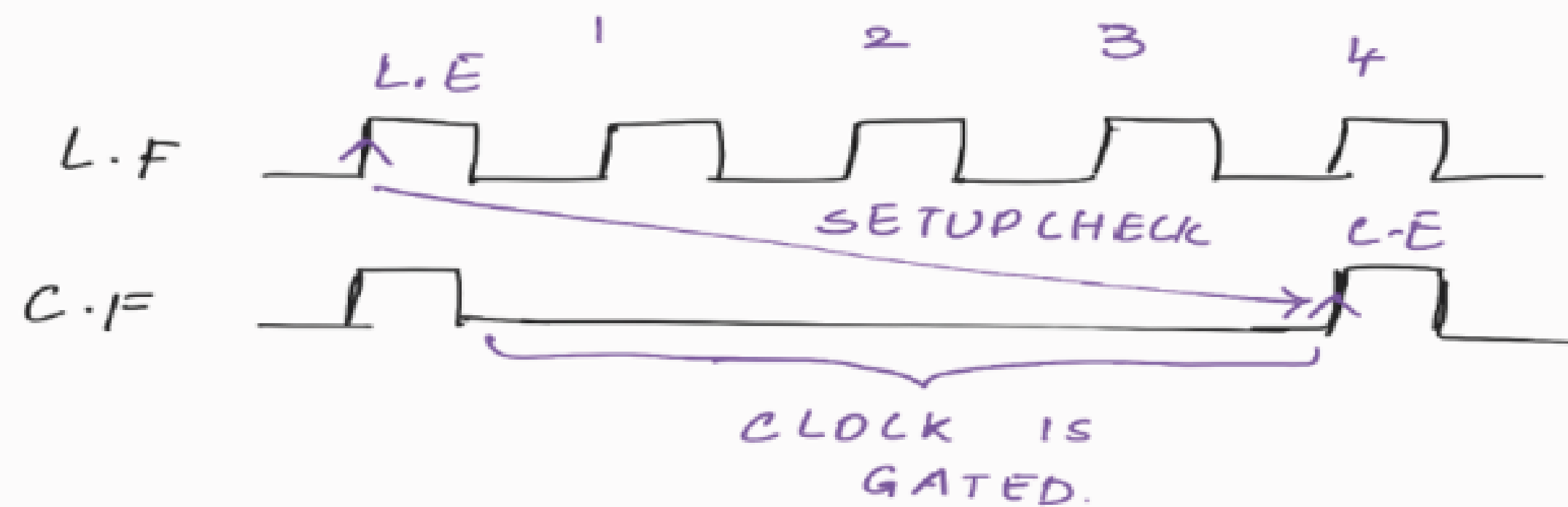


the data at the i/p of capture ff will be stabilized only after 4 cycles. So designer has given MCP.

only if $EN=1$, clock will be allowed to the receiver and only after that receiver will sample the data at its i/p.

When data at the i/p of capture flop is stabilized, En will be made 1.

Until $En=1$, clock to the receiver will be gated as shown:



set-multicycle-path -setup 4 -from f+1/Q to f+2/D
↳ Given in SDC.

In DFT, we don't cover TDF for this scenario, because in TDF, while generating patterns, we have to give pulses like this :- 

not $\rightarrow$ 

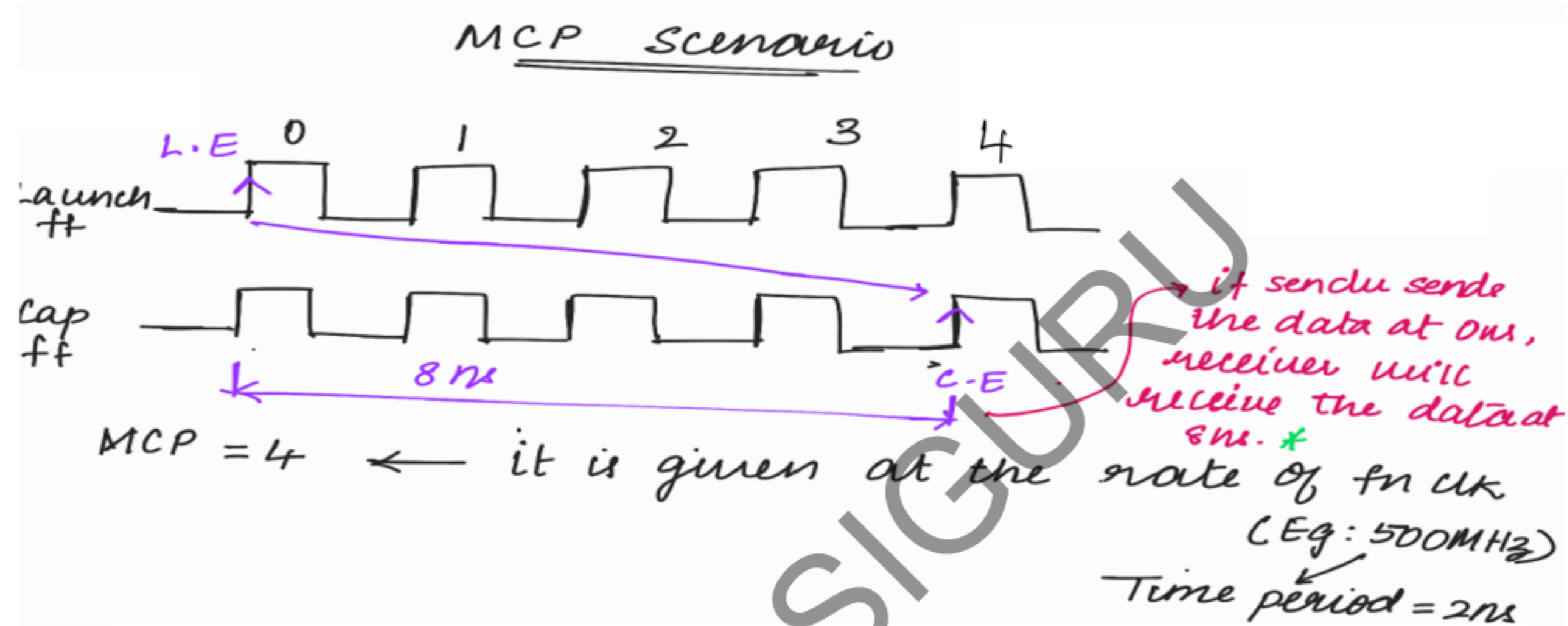
If we give the clock like this , timing will fail for that path because T_{combo} is big and data at the i/p of capture flop wouldn't have been stabilized during this period.

Even if suppose we bring the clock pulses as shown below:

, then the other paths will be under tested.

Because for all the other paths we have to test whether the transition of data happens within  this period.

How MCP is not a problem during SA?



For SA, we have the slow freq clock.

SCLK is in the range of 40-80 MHz. So by the time the clock edge of SCLK comes, the receiver will be capable of sampling the data.

∴ This is not a problem.

$$f = 40 \text{ MHz} \Rightarrow T_p = 25 \text{ ns}$$



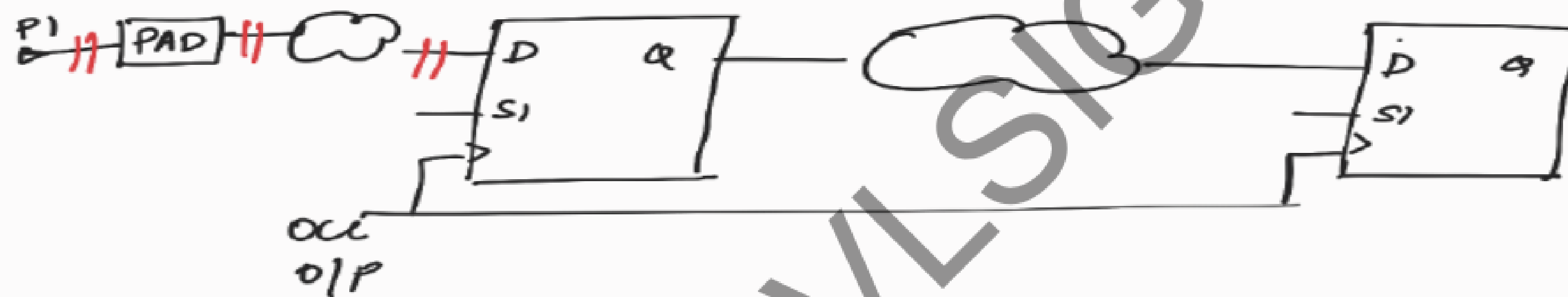
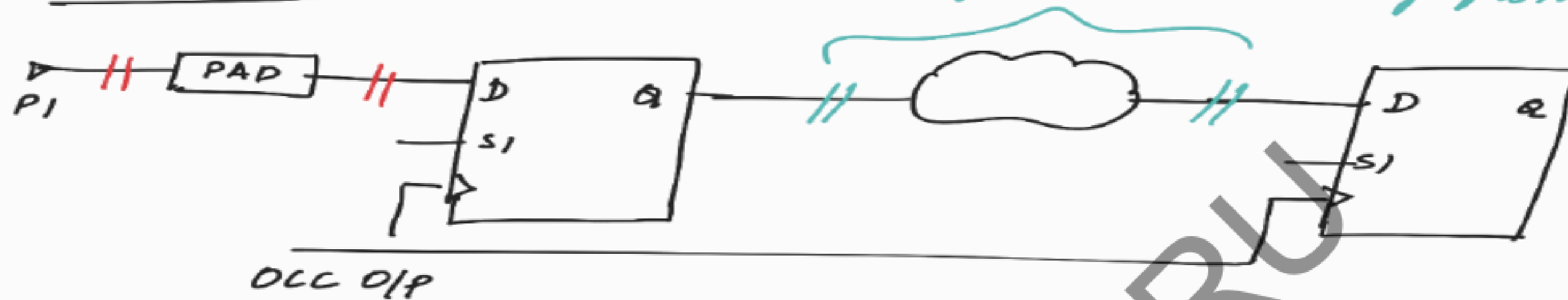
$$f = 80 \text{ MHz} \Rightarrow T_p = 12.5 \text{ ns}$$



REASON 3 :

Reason 3 :

tool will be able to make a transition as the clocks of ffs are coming from PLL through OLC



// $\Rightarrow$ in these nets, our tool will not bring a transition.

Inside the chip, we have high freq TDF domain (Eg: 500MHz - 1GHz)

REASON 3 (Contd...) :

But PADs don't support such high data rate

So tool holds PI to a constant 0 or 1 during capture phase. So transition can't be launched from PI.

AU.HPI → Faults in the fanout cone of Hold PIs.